PETER A. SAYEGH

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EDUCATION

Columbia University — New York, NY
MS in Electrical Engineering – Nikola Tesla Scholarship, GPA: 3.80/4.0 — Expected Dec 2026
Coursework: Digital Signal Processing, Adv. Analog Integrated Circuits (IC), Power Electronics, MOS Transistors, Analog-Digital Interfaces in VLSI, Embedded AI, Microwave Circuit Design, Power Management IC, Millimeter-wave IC Design
Activities: Lead Department Representative for Electrical Engineering within the Engineering Student Graduate Council (EGSC)

École Polytechnique — Palaiseau, FR
BS in Mathematics and Physics, GPA: 3.75/4.0 — Jun 2025
Coursework: Convex Optimization, Stochastic Processes, Electrodynamics, Adv. Quantum Physics, Thermodynamics, Statistical Physics, Condensed Matter, Algorithms, Numerical Analysis, Web Programming, Object-Oriented Programming, Machine Learning

EXPERIENCE

Illumination Diagnostics — New York, NY
Electrical Engineering Intern — May 2026 - Present

- Developed a heart rate extraction pipeline for a 12-channel optical wearable, achieving 1-3 bpm clinical accuracy
- Prototyped PPG and temperature acquisition systems using ESP32 microcontrollers with synchronized respiration belt recording
- Assembled an IR LED illumination array on a perfboard to provide stable background illumination, reducing ambient light interference in pupil tracking

TriSpan LLP — New York, NY
Private Equity Intern — Jul - Aug 2024

- Streamlined Profit & Loss reporting across 10 portfolio companies using Excel
- Conducted Leveraged Buyout modeling using Excel and Bloomberg terminals
- Drafted valuation reports using Discounted Cash Flow analysis for portfolio companies worth $200M+ in assets

ELECTRICAL ENGINEERING PROJECTS

Columbia University — New York, NY
Fully Differential Switched-Capacitor Amplifier — Feb – May 2026

- Designed folded-cascode OTA with CMFB in 0.18 µm CMOS, achieving 56.6 dB gain, 71° phase margin, and 333 dB CMRR at 486 µW
- Derived settling and slew-rate constraints to meet 0.09% settling error within 28 ns for a 7-bit 16 MHz pipelined ADC

Columbia University — New York, NY
SignSpeak: ASL Recognition on Microcontroller — Feb – May 2026

- Deployed 16K-parameter depthwise-separable CNN on ESP32S3 via TFLite Micro INT8 quantization, achieving 97.4% accuracy in 67.4 KB flash
- Built a full on-device inference pipeline from camera capture to OLED display, <110 ms latency with no cloud dependency

Columbia University — New York, NY
250 MS/s 8-bit Current-Steering DAC — Feb - May 2026

- Designed 3+5 segmented current-steering DAC in 0.18 µm CMOS with thermometer MSBs and per-cell trim DACs, achieving SFDR >66 dB and DNL $<2\times10^{-4}$ LSB across all process corners
- Optimized cascode unit cell sizing to meet 1% mismatch budget, delivering ≥1.53 $V_{pp,diff}$ into 100 Ω at 250 MS/s within 27.6 mW

École Polytechnique - Center for Theoretical Physics — Palaiseau, FR
Dynamic Stability and Levitation Control in Magnetic Levitation (Maglev) Systems — Feb 2024 - Jul 2025

- Designed and constructed a scaled Maglev train prototype using magnetic field analysis, achieving 96% theoretical validation
- Extrapolated results to a real Electromagnetic Suspension engine scale, predicting capacity within 2% of Shanghai Maglev

TECHNICAL SKILLS

CAD Tools: Cadence Spectre, PLECS, LTSpice, Keysight ADS, Xschem, ngspice, Magic VLSI, KLayout, OpenROAD
Programming Languages: Python, C++, Arduino, R, MATLAB, HTML, CSS, JavaScript, QML
Data Science and Machine Learning: TensorFlow, Keras, JAX/Optax, scikit-learn, NumPy, SciPy, pandas, statsmodels, DSPy
Scientific Computing: QuTiP, Scikit-HEP, PySpice
Languages: French (Native), Arabic (Native), German (Advanced), Spanish (Intermediate)